

PocketSmart AI

Your Smart Budget & Recommendation Assistant

Project Description:

PocketSmartAI is a smart budget and recommendation assistant designed to help users make personalized decisions based on their budget, preferences, and requirements. The application uses AI-based recommendations to provide suitable suggestions for different planning needs.

The system includes three major planning modules: Home Interior Planner, Party Planner, and Jewelry Planner. The Home Interior Planner considers the user's budget, selected rooms, lights, fans, furniture, dining requirements, and additional preferences. The Party Planner considers budget, number of guests, party type, venue, catering, decoration, and entertainment requirements. The Jewelry Planner considers budget, occasion, style, and optionally an outfit image to generate suitable recommendations.

The application uses a FastAPI backend with Python, a web-based frontend using HTML, CSS, JavaScript and Jinja2, and Google Gemini for AI-powered recommendation generation. Authentication is implemented using JWT with Passlib/Bcrypt, while Pillow is used for image-related processing. The application also includes fallback logic for recommendation generation when required.

Scenarios

Scenario 1: Home Interior Planning

A user wants to plan their home interior within a specific budget. They enter the total budget, number of lights, fans, furniture and dining tables, select the required rooms such as living room, kitchen and bedroom, and provide additional requirements. PocketSmartAI processes these inputs and generates suitable budget-based interior recommendations.

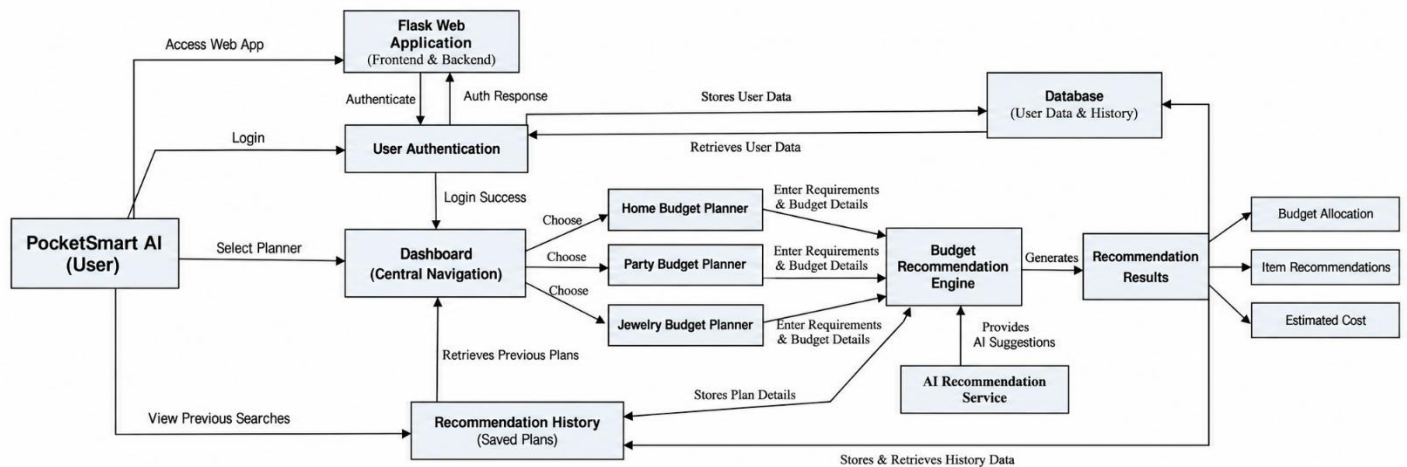
Scenario 2: Party Planning

A user wants to organize a party within a fixed budget. They provide the number of guests, party type, venue, catering, decoration and entertainment requirements. PocketSmartAI analyzes these requirements and provides recommendations suitable for the available budget.

Scenario 3: Jewelry Planning

A user wants to select jewelry for a particular occasion. They provide their budget, occasion and preferred style. They can also provide an outfit image. PocketSmartAI uses these inputs to generate suitable jewelry recommendations based on their requirements.

Technical Architecture:



Architecture Flow

User → Web Interface → FastAPI Backend → Input Validation → Planner Module → Google Gemini → Recommendation Response → User Interface

Pre-requisites:

- Python programming knowledge
- FastAPI framework knowledge
- HTML, CSS and JavaScript knowledge
- Jinja2 template knowledge
- Google Gemini API familiarity
- REST API knowledge
- JWT authentication concepts
- Basic image processing knowledge using Pillow
- Git and GitHub knowledge
- Playwright testing knowledge
- VS Code / Python development environment

Project Workflow:

Milestone 1: Project Setup and Architecture

Activity 1.1: Set up the PocketSmartAI development environment.

Activity 1.2: Configure the FastAPI backend and required Python dependencies.

Activity 1.3: Define the frontend, backend and AI recommendation architecture.

Activity 1.4: Configure Google Gemini integration and required environment variables.

Milestone 2: Authentication Development

Activity 2.1: Develop user registration functionality.

Activity 2.2: Implement user login functionality.

Milestone 3: Planner Module Development

Activity 3.1: Home Interior Planner

Develop the Home Interior Planner to accept:

- Total budget
- Number of lights
- Number of fans
- Number of furniture items
- Number of dining tables
- Living room selection
- Kitchen selection
- Bedroom selection
- Additional requirements

The backend processes these inputs and generates suitable home interior recommendations.

Milestone 4: AI Recommendation Integration

Activity 4.1: Google Gemini Integration

Integrate Google Gemini with the FastAPI backend to process planner inputs and generate personalized recommendations.

Activity 4.3: Recommendation Processing

Process the AI response and return the generated recommendations to the frontend in a user-friendly format.

Milestone 5: Frontend Development

Activity 5.1: Form Handling

Create user input forms for each planner and send the collected information to the FastAPI backend.

Activity 5.2: Recommendation Display

Display AI-generated recommendations clearly after successful processing.

Milestone 1: AI Model Selection and Application Architecture

In this milestone, the focus is on selecting and configuring the appropriate AI technology for PocketSmartAI's personalized recommendation requirements. The application uses Google Gemini to generate recommendations based on the user's budget, preferences, requirements, and selected planning category.

The architecture is designed to connect the web interface with the FastAPI backend, AI recommendation layer, authentication system, and individual planning modules.

Activity 1.1: Configure Google Gemini AI

- Create and configure the required Google Gemini API access.
- Store the API configuration securely using environment variables.
- Integrate the google-genai package with the Python backend.
- Configure the AI service to process user requirements and generate personalized recommendations.
- Implement a deterministic local fallback for recommendation generation when AI generation is unavailable.

Activity 1.2: Define Recommendation Requirements

Identify the information required by each PocketSmartAI module:

Home Interior Planner

- Total budget
- Number of lights
- Number of fans
- Furniture requirements
- Dining table requirements
- Required rooms
- Additional requirements

Party Planner

- Budget
- Number of guests
- Party type
- Venue
- Catering
- Decoration
- Entertainment

Jewelry Planner

- Budget
- Occasion
- Jewelry style
- Optional outfit image

The collected information is used to generate recommendations relevant to the user's requirements.

Activity 1.3: Define Application Architecture

The PocketSmartAI architecture consists of:

Frontend → FastAPI Backend → Authentication / Planner Module → Gemini AI → Recommendation Response → Frontend

Activity 1.4: Set Up Development Environment

- Install Python and required dependencies.
- Configure FastAPI and Uvicorn.
- Configure Jinja2 templates.
- Set up Google Gemini integration.
- Configure JWT authentication.
- Configure Passlib/Bcrypt for password security.
- Configure Pillow for image processing.
- Prepare the project structure for planner modules.
- Configure Playwright for end-to-end testing.

Milestone 2: Core Functionality Development

This milestone focuses on implementing the major planning modules of PocketSmartAI and connecting them with the AI recommendation system.

Activity 2.1: Develop Home Interior Planner

Develop the Home Interior Planner to collect the user's:

- Budget
- Lights
- Fans
- Furniture
- Dining table requirements
- Living room requirement
- Kitchen requirement
- Bedroom requirement
- Additional requirements

The collected information is processed by the backend and used to generate suitable interior recommendations.

Activity 2.2: Develop Party Planner

Develop the Party Planner to collect:

- Budget
- Number of guests
- Party type
- Venue
- Catering requirements
- Decoration requirements
- Entertainment requirements

The system processes these details and generates recommendations according to the user's requirements and available budget.

Activity 2.3: Develop Jewelry Planner

Develop the Jewelry Planner to collect:

- Budget
- Occasion
- Jewelry style
- Optional outfit image

The uploaded outfit image can be processed using Pillow, while the remaining user requirements are passed to the recommendation workflow.

Activity 2.4: Implement AI Recommendation Generation

- Collect planner-specific user inputs.
- Construct a structured recommendation request.
- Send the relevant information to Google Gemini.
- Process the generated response.
- Display the recommendation to the user.
- Use the local fallback mechanism when required.

Milestone 3: FastAPI Backend Development

Milestone 3 focuses on developing the backend logic that controls authentication, planner requests, AI recommendation generation, and response handling.

Activity 3.1: Develop FastAPI Application

Create the FastAPI application to handle:

- User authentication
- Planner requests
- Recommendation generation
- Image-related requests
- API responses

Uvicorn is used as the application server for running the FastAPI application.

Activity 3.2: Implement User Authentication

Implement authentication functionality including:

- User registration
- User login
- Password validation
- JWT-based authentication
- Secure password hashing using Passlib/Bcrypt
- Protected application access

Activity 3.3: Implement Planner API Logic

Create backend logic for the three planner modules:

- Home Interior Planner
- Party Planner
- Jewelry Planner

Each module accepts its specific requirements and passes the information to the recommendation workflow.

Activity 3.4: Implement Recommendation Response Handling

The backend receives the AI-generated recommendation, processes the response and sends structured information back to the frontend.

Error handling is implemented to manage invalid inputs, failed AI responses and other application-level errors.

Milestone 4: Frontend Development

Milestone 4 focuses on developing the user interface of PocketSmartAI and providing a simple workflow for users to enter their requirements and receive recommendations.

Activity 4.1: Design and Develop User Interface

Develop the frontend using:

- HTML
- CSS
- JavaScript
- Jinja2

Create interfaces for:

- Home page
- Registration
- Login
- Dashboard
- Home Interior Planner
- Party Planner
- Jewelry Planner
- Recommendation results

Activity 4.2: Create Planner Forms

Develop individual forms for each planning module.

Users can enter their requirements through structured input fields and submit the information to the FastAPI backend.

Activity 4.3: Display AI Recommendations

After processing the request:

User Input → Backend Processing → AI Recommendation → Result Display

The generated recommendation is displayed clearly on the corresponding planner result page.

Activity 4.4: Responsive Interface

The frontend is designed to work across different screen sizes, including:

- Desktop
- Laptop
- Tablet
- Mobile

Milestone 5: Testing and Deployment

Milestone 5 focuses on validating PocketSmartAI's complete workflow and ensuring that the application's major features function correctly.

Activity 5.1: Backend Testing

Test the FastAPI backend to verify:

- API availability
- Request handling
- Input validation
- Authentication
- Planner processing
- Recommendation generation
- Error handling

Activity 5.2: Authentication Testing

Verify the complete authentication flow:

Registration → Login → Authentication → Dashboard → Planner Access

Test valid and invalid credentials to ensure the authentication system works correctly.

Activity 5.3: Planner Testing

Test each planner independently.

Home Interior Planner

Verify budget, room, furniture, lights, fans and additional requirement inputs.

Party Planner

Verify guest count, party type, venue, catering, decoration and entertainment inputs.

Jewelry Planner

Verify budget, occasion, style and optional outfit image processing.

Activity 5.4: Playwright End-to-End Testing

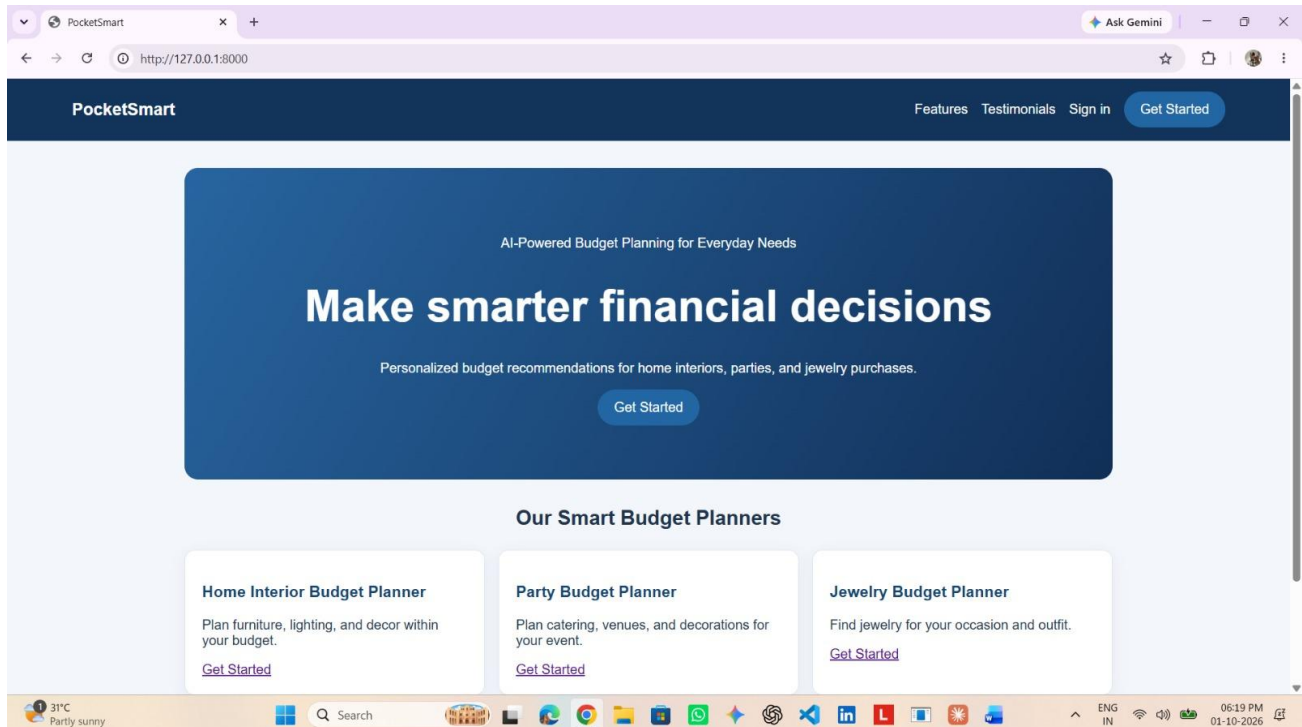
Use Playwright to test the complete user journey through the browser.

The E2E testing workflow verifies important application flows such as:

- Opening the application
- Registration/login
- Dashboard navigation
- Planner selection
- Form submission
- Recommendation generation
- Result display

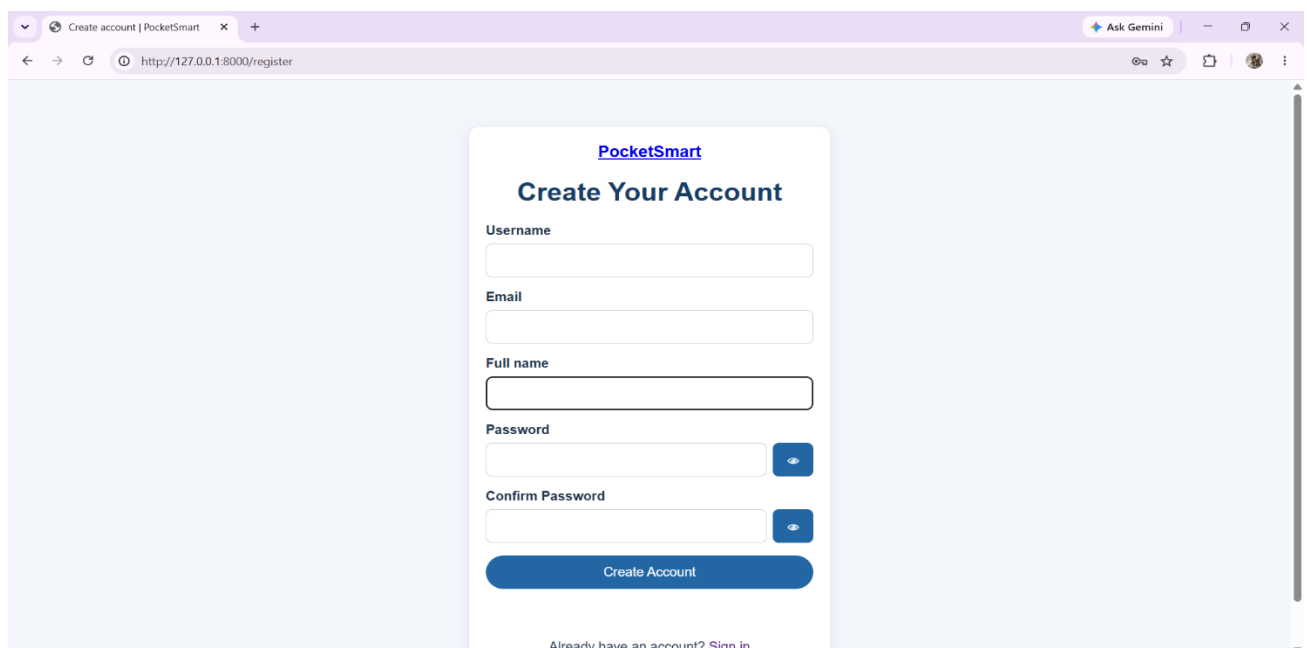
Activity 5.5: Final Verification

Perform final verification of the complete application:



Home Page / Landing Page – Description

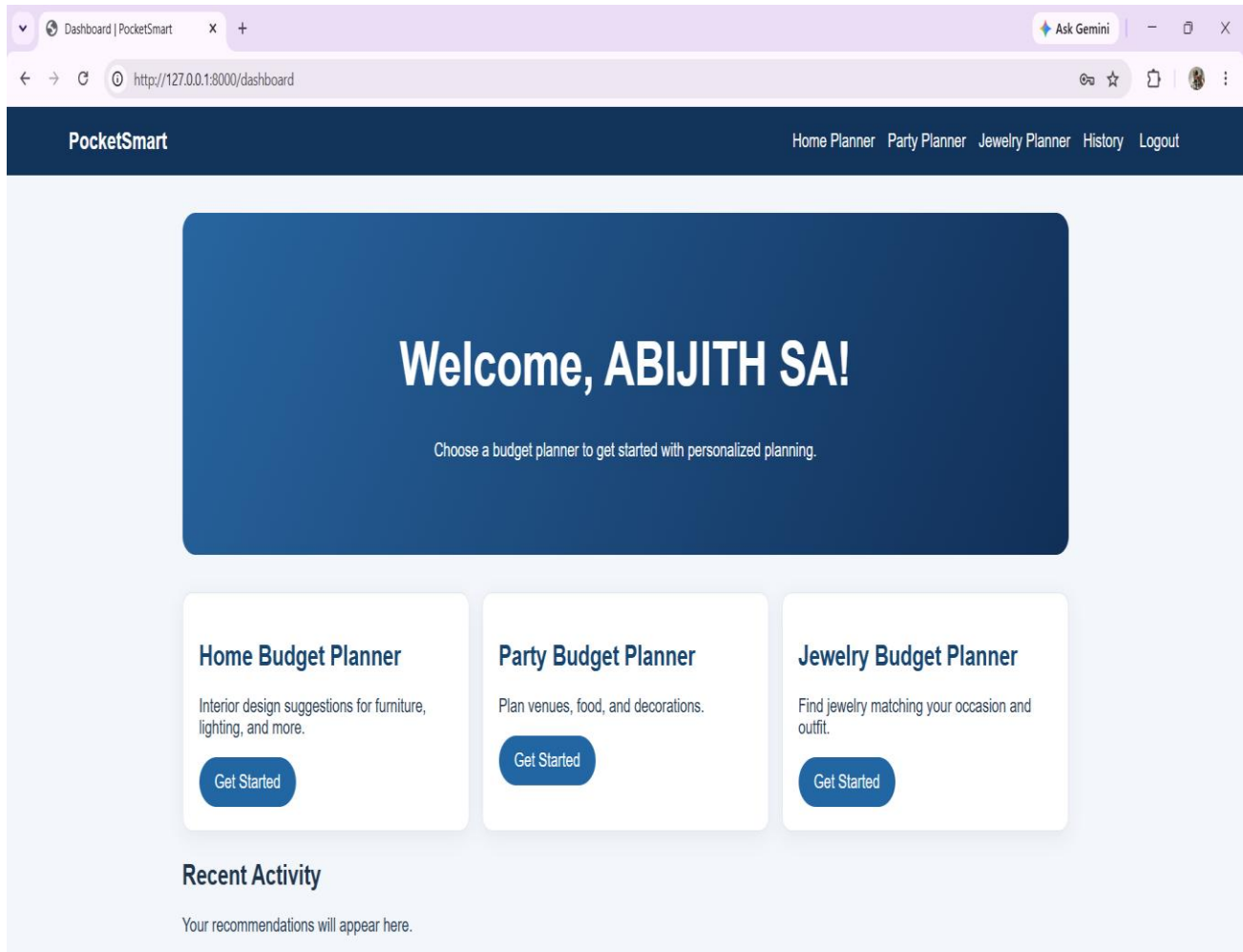
Description: The PocketSmartAI home page introduces the platform as an AI-powered budget planning assistant. It provides users with personalized budget recommendations for **home interiors, parties, and jewelry purchases**, along with easy navigation options such as **Features, Testimonials, Sign in, and Get Started**. The page also highlights the three main budget planning modules available in the application.



Create Account / Registration Page – Description

Description:

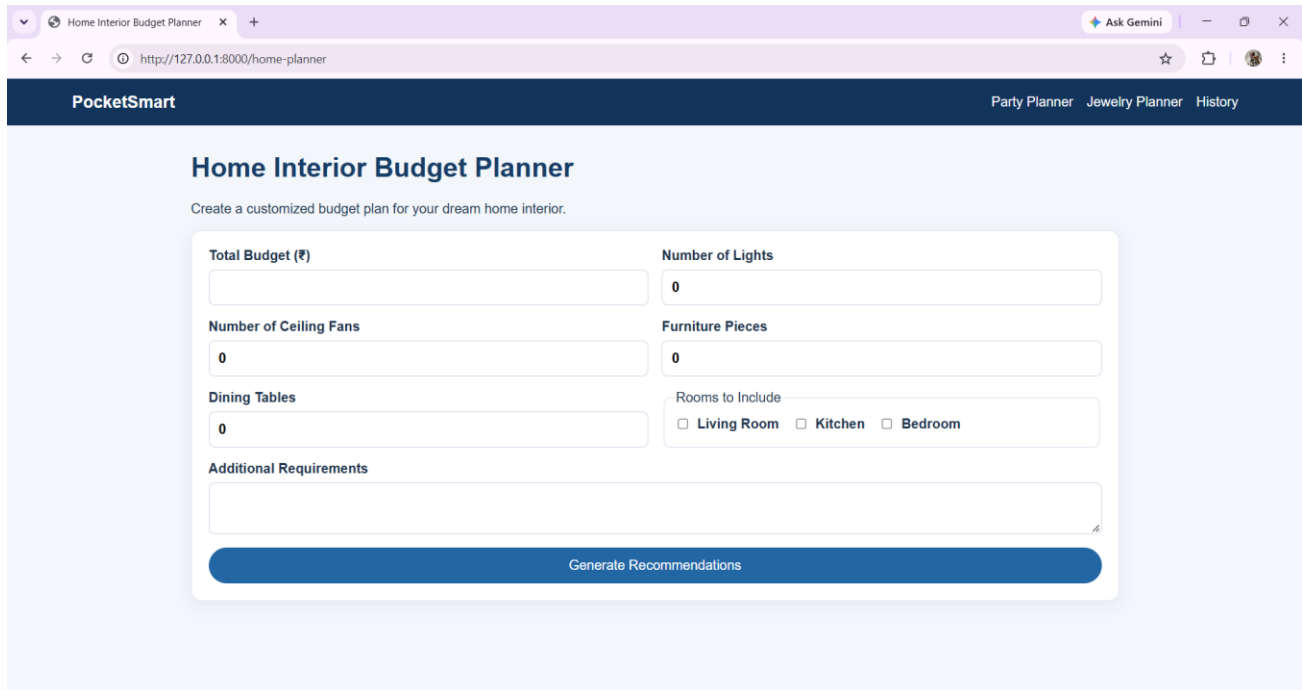
The Registration page allows new users to create a PocketSmartAI account by providing their username, email, full name, and password. It includes password confirmation and show/hide password options to improve usability and account security. After entering the required details, users can select **Create Account** to register and access the PocketSmartAI platform



Dashboard Page – Description

Description:

The PocketSmartAI Dashboard welcomes the authenticated user and provides access to the three main budget planning modules: **Home Budget Planner, Party Budget Planner, and Jewelry Budget Planner**. Users can select a planner based on their requirements and access their previous recommendations through the **History** section. The dashboard also provides navigation options for Home, Planners, History, and Logout.



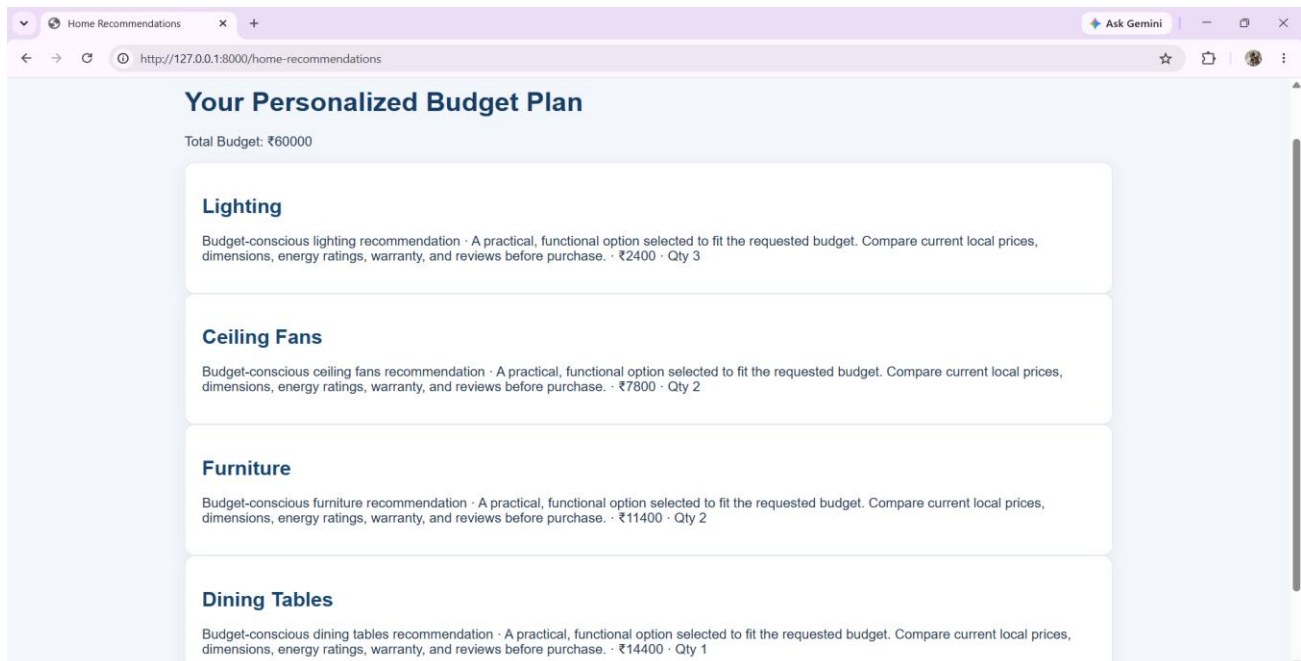
The screenshot shows a web browser window with the URL `http://127.0.0.1:8000/home-planner`. The page title is "Home Interior Budget Planner". Below the title, it says "Create a customized budget plan for your dream home interior." The form contains several input fields and checkboxes:

- Total Budget (₹)**: A text input field.
- Number of Lights**: A text input field with the value "0".
- Number of Ceiling Fans**: A text input field with the value "0".
- Furniture Pieces**: A text input field with the value "0".
- Dining Tables**: A text input field with the value "0".
- Rooms to Include**: A section with three checkboxes: ☐ Living Room, ☐ Kitchen, and ☐ Bedroom.
- Additional Requirements**: A large text area for additional input.
- Generate Recommendations**: A blue button at the bottom of the form.

Home Interior Budget Planner – Description

Description:

The Home Interior Budget Planner allows users to create a customized interior budget plan by entering their **total budget, number of lights, ceiling fans, furniture pieces, and dining tables**. Users can also select the required rooms, such as **Living Room, Kitchen, and Bedroom**, and provide additional requirements. The entered details are processed to generate personalized home interior recommendations.



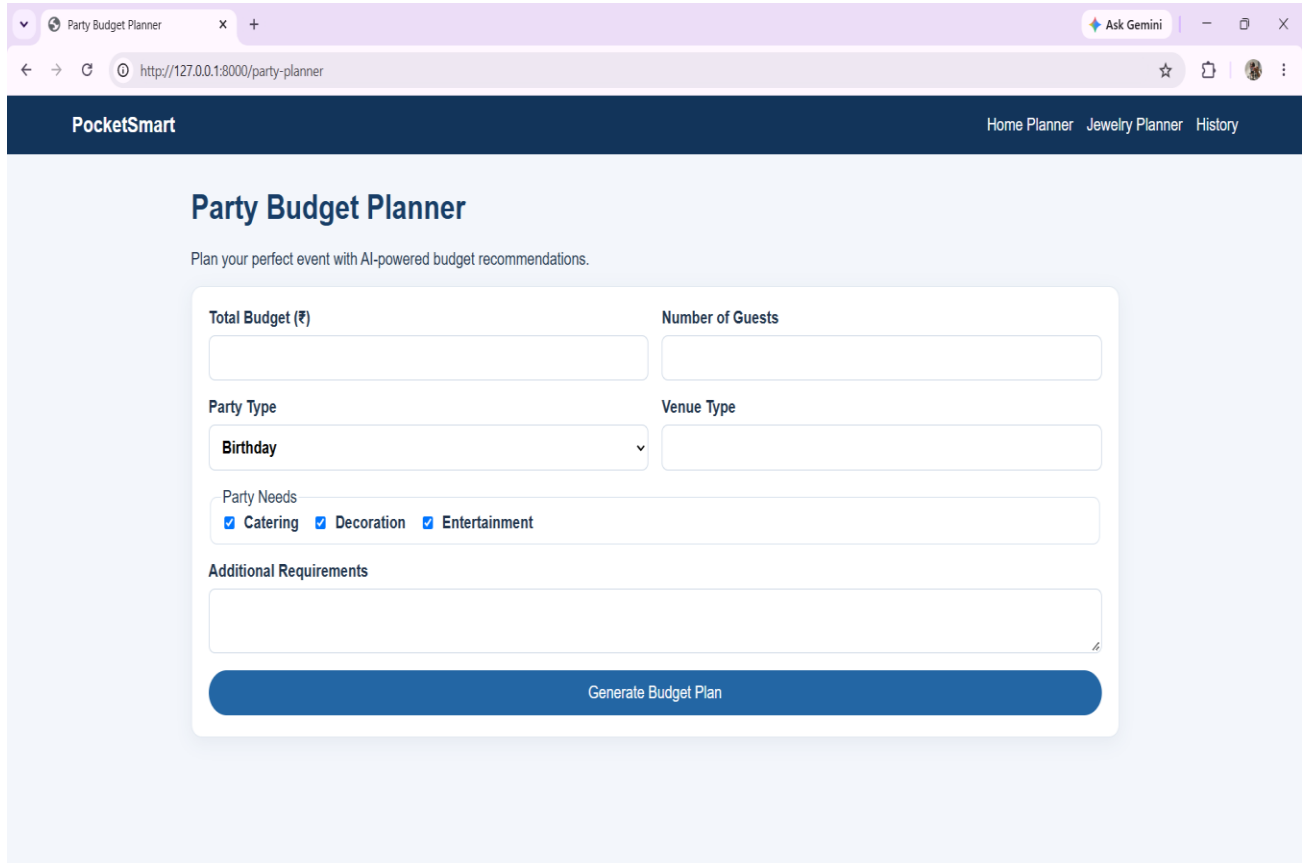
The screenshot shows a web browser window with the URL `http://127.0.0.1:8000/home-recommendations`. The page title is "Your Personalized Budget Plan". Below the title, it says "Total Budget: ₹60000". The page displays four recommendation cards:

- Lighting**: Budget-conscious lighting recommendation · A practical, functional option selected to fit the requested budget. Compare current local prices, dimensions, energy ratings, warranty, and reviews before purchase. · ₹2400 · Qty 3
- Ceiling Fans**: Budget-conscious ceiling fans recommendation · A practical, functional option selected to fit the requested budget. Compare current local prices, dimensions, energy ratings, warranty, and reviews before purchase. · ₹7800 · Qty 2
- Furniture**: Budget-conscious furniture recommendation · A practical, functional option selected to fit the requested budget. Compare current local prices, dimensions, energy ratings, warranty, and reviews before purchase. · ₹11400 · Qty 2
- Dining Tables**: Budget-conscious dining tables recommendation · A practical, functional option selected to fit the requested budget. Compare current local prices, dimensions, energy ratings, warranty, and reviews before purchase. · ₹14400 · Qty 1

Home Recommendations Page – Description

Description:

The Home Recommendations page displays a personalized budget plan based on the user's selected requirements and total budget. It provides budget-conscious recommendations for **lighting, ceiling fans, furniture, and dining tables**, along with the suggested price and quantity for each item. This helps users plan their home interior purchases while staying within their specified budget.



The screenshot shows a web browser window with the title 'Party Budget Planner'. The address bar shows the URL 'http://127.0.0.1:8000/party-planner'. The page has a dark blue header with the 'PocketSmart' logo on the left and navigation links 'Home Planner', 'Jewelry Planner', and 'History' on the right. The main content area is light blue and features the title 'Party Budget Planner' and the subtitle 'Plan your perfect event with AI-powered budget recommendations.' Below this is a white form with the following fields:

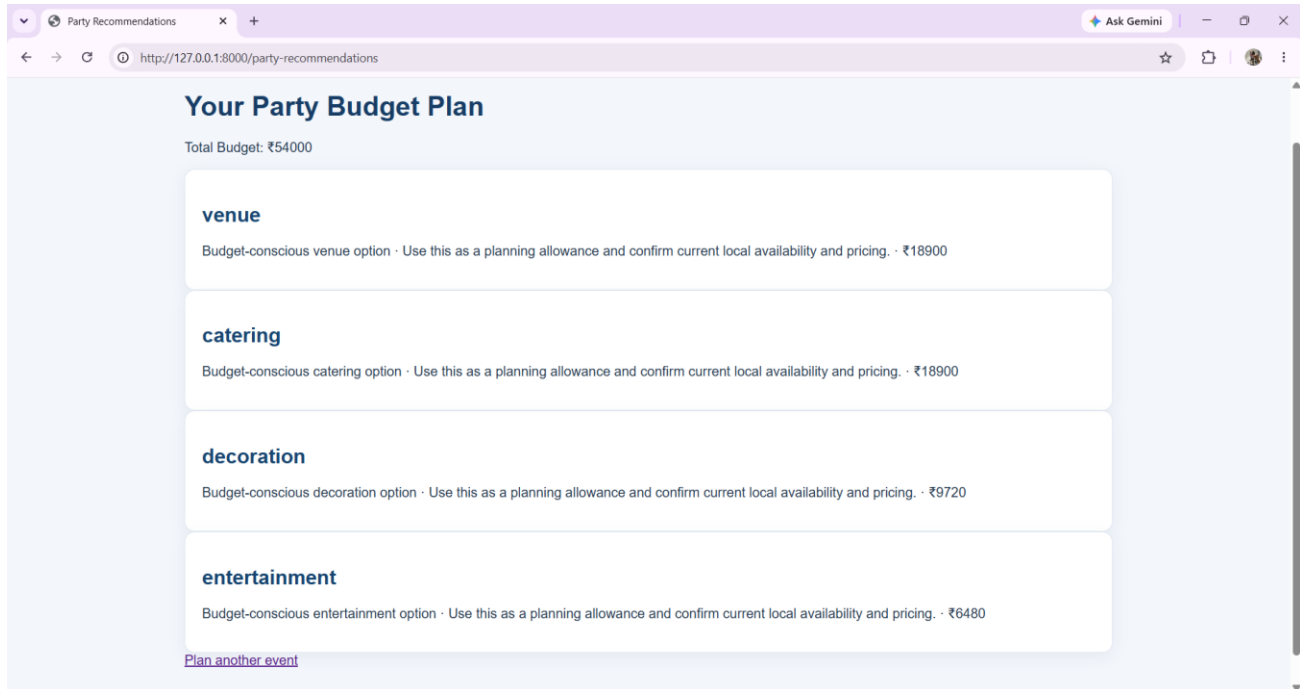
- Total Budget (₹)**: A text input field.
- Number of Guests**: A text input field.
- Party Type**: A dropdown menu with 'Birthday' selected.
- Venue Type**: A text input field.
- Party Needs**: A row of three checkboxes: 'Catering' (checked), 'Decoration' (checked), and 'Entertainment' (checked).
- Additional Requirements**: A large text area for additional input.

At the bottom of the form is a blue button labeled 'Generate Budget Plan'.

Party Budget Planner – Description

Description:

The Party Budget Planner allows users to create a customized event budget plan by entering the **total budget, number of guests, party type, and venue type**. Users can select required services such as **catering, decoration, and entertainment**, along with additional requirements. The entered details are processed to generate AI-powered budget recommendations for the event.



Party Recommendations

Ask Gemini

http://127.0.0.1:8000/party-recommendations

Your Party Budget Plan

Total Budget: ₹54000

venue

Budget-conscious venue option · Use this as a planning allowance and confirm current local availability and pricing. · ₹18900

catering

Budget-conscious catering option · Use this as a planning allowance and confirm current local availability and pricing. · ₹18900

decoration

Budget-conscious decoration option · Use this as a planning allowance and confirm current local availability and pricing. · ₹9720

entertainment

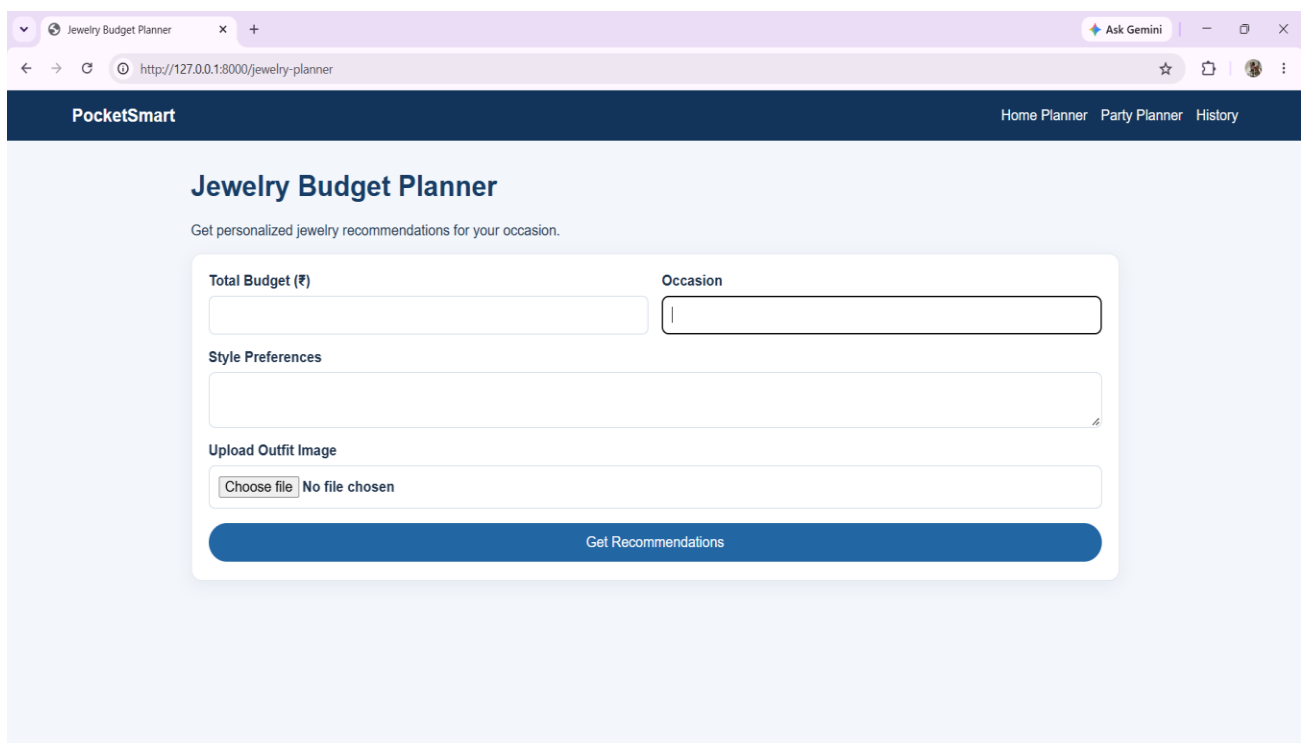
Budget-conscious entertainment option · Use this as a planning allowance and confirm current local availability and pricing. · ₹6480

[Plan another event](#)

Party Recommendations Page – Description

Description:

The Party Recommendations page displays a personalized budget plan based on the user's selected event requirements and total budget. It provides budget allocations for **venue, catering, decoration, and entertainment**, helping users organize the event while keeping the planned expenses within the specified budget.



Jewelry Budget Planner

Get personalized jewelry recommendations for your occasion.

Total Budget (₹)

Occasion

Style Preferences

Upload Outfit Image

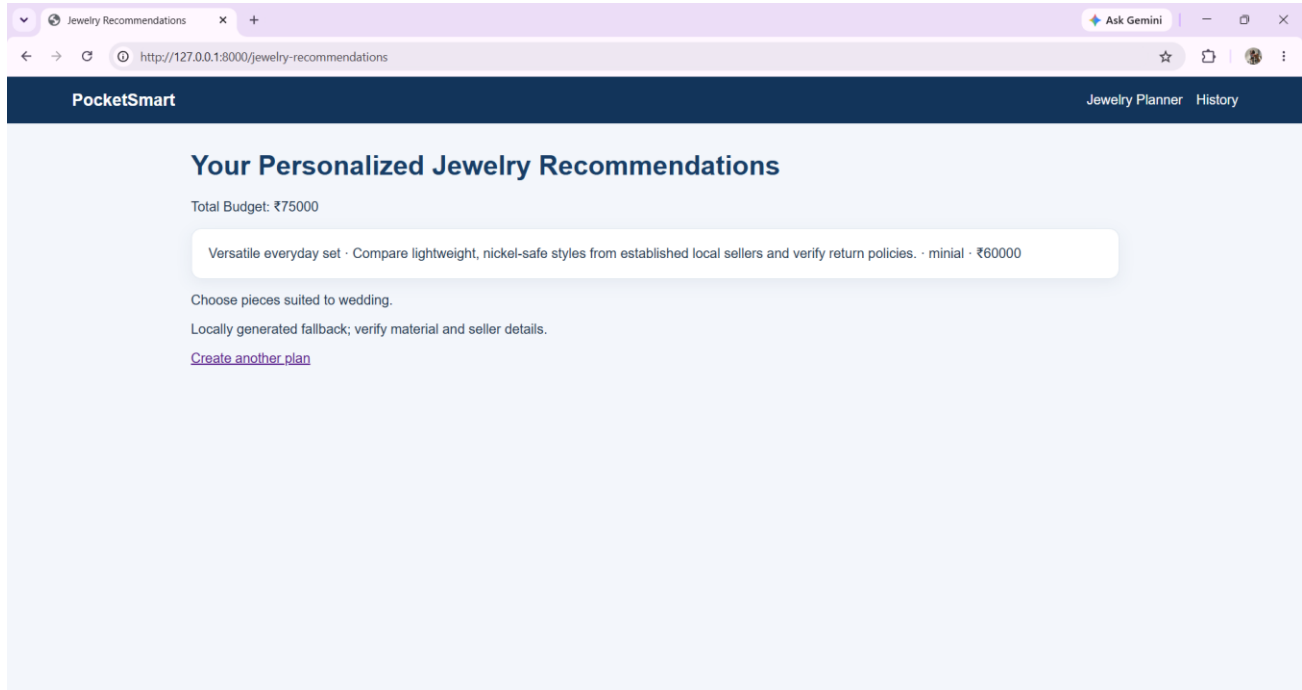
Choose file No file chosen

Get Recommendations

Jewelry Budget Planner – Description

Description:

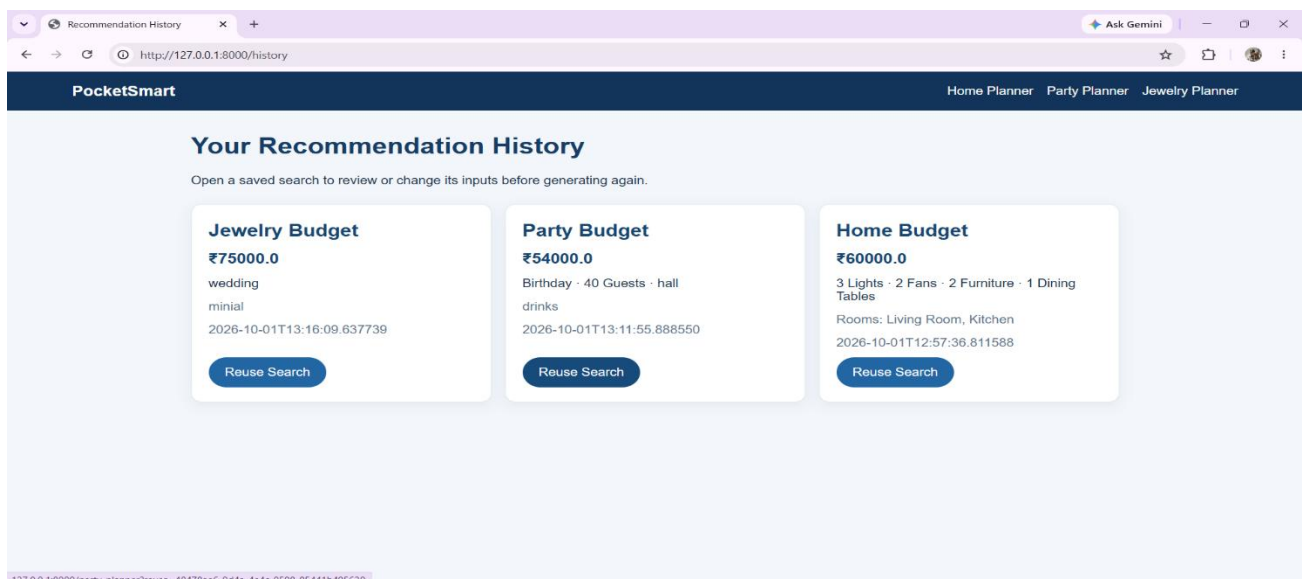
The Jewelry Budget Planner allows users to enter their **total budget, occasion, and preferred jewelry style** to receive personalized jewelry recommendations. Users can also upload an **outfit image** to help match the recommendations with their outfit. The entered details and image are processed to generate suitable jewelry suggestions based on the user's requirements.



Jewelry Recommendation Result

Description:

The system displays personalized jewelry recommendations based on the user's **budget, occasion, and style preferences**. It provides a suitable jewelry suggestion along with the estimated cost and guidance to verify material and seller details before purchase.



Recommendation History

Description:

The Recommendation History page stores previously generated **Home, Party, and Jewelry budget plans**. Users can review their saved inputs, budget details, and preferences, and use “**Reuse Search**” to reopen a previous plan and generate recommendations again.